



Giant density fluctuations in locally hyperuniform states

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曹溪能

1. 背景介绍

- random organization model
- 向列型vicsek model

2. 模型介绍

3. 模拟结果

4. 流体力学理论

5. 另一种现象

6. 总结

背景介绍: random organization model

- 运动规则: 重叠 $\longrightarrow$ 随机弹开
- 有活性态-吸收态转变
- 粒子守恒情况下, 相变属于CDP (守恒定向渗流普适类)

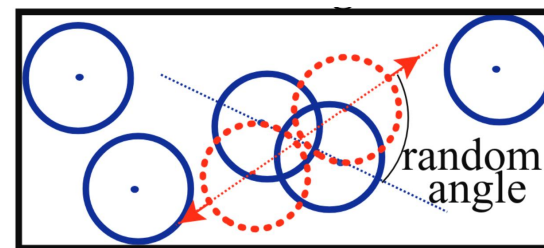


图1

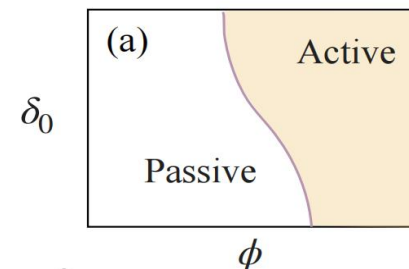


图2

- 场方程

$$\partial_t \rho(\mathbf{x}, t) = D_\rho \nabla^2 \rho(\mathbf{x}, t) + (a - \lambda \phi(\mathbf{x}, t)) \rho(\mathbf{x}, t) - b \rho(\mathbf{x}, t)^2 + \sqrt{\rho(\mathbf{x}, t)} \eta(\mathbf{x}, t),$$

$$\partial_t \phi(\mathbf{x}, t) = \mu \nabla^2 \rho(\mathbf{x}, t).$$

ρ 是活性密度场, ϕ 是总密度场

- 超均匀性

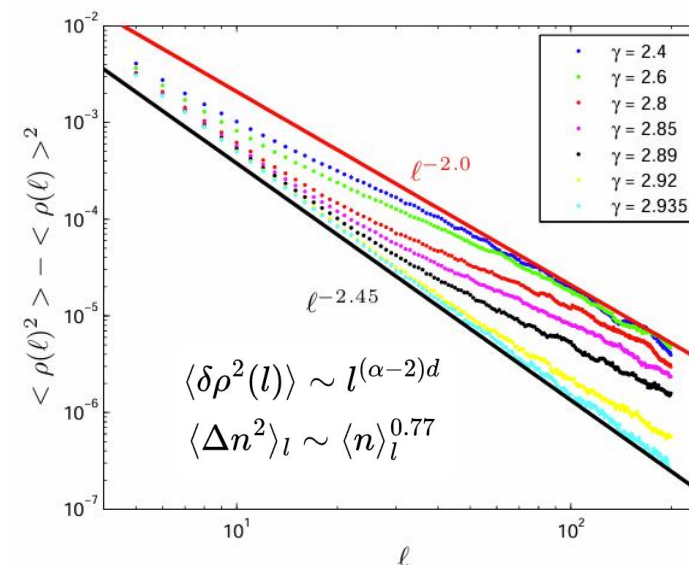


图3

Hexner D, Levine D. Hyperuniformity of critical absorbing states[J]. Physical review letters, 2015, 114(11): 110602.

背景介绍: 向列型vicsek model

• **运动规则** $\theta_j^{t+\Delta t} = \frac{1}{2} \text{Arg} \left[\sum_{k \in V_j} e^{i2\theta_k^t} \right] + \psi_j^t$

$$\mathbf{r}_i^{t+1} = \mathbf{r}_i^t \pm \delta_0 \mathbf{n}_i^t$$

• **场方程** $\nabla = \partial_x + i\partial_y, \Delta = \nabla \nabla^*$

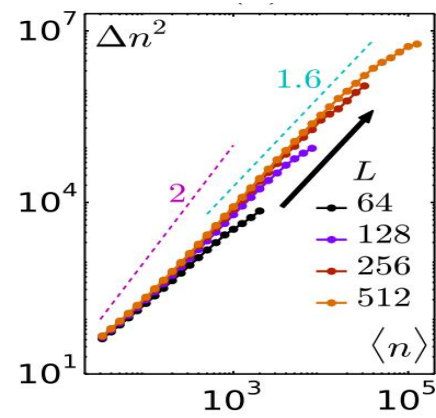
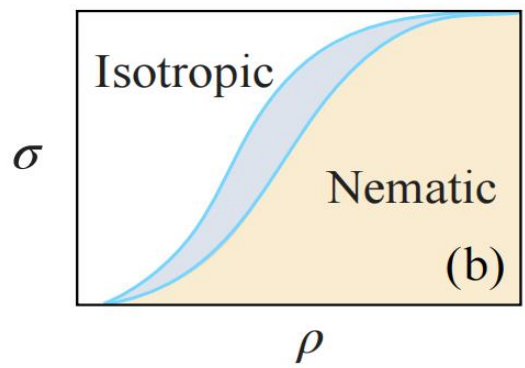
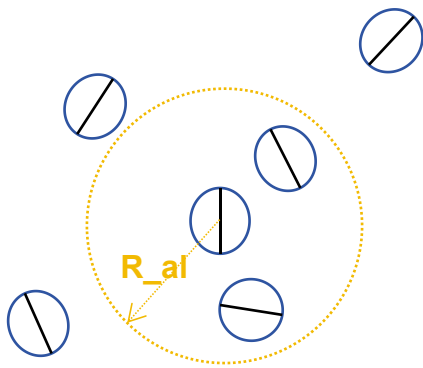
$$\partial_t \rho = \frac{1}{2} \Delta \rho + \frac{1}{2} \text{Re}(\nabla^{*2} Q),$$

$$\partial_t Q = (\mu(\rho) - \xi |Q|^2) Q + \frac{1}{4} \nabla^2 \rho + \frac{1}{2} \Delta Q,$$

• 向列有序相存在巨涨落

线性理论: $\langle \Delta n^2 \rangle_l \sim \langle n \rangle_l^2$

• 和vicsek模型的区别



D. Hexner and D. Levine,
Phys. Rev. Lett. **114**, 110602
(2015).

Class	Basic features			Coexistence phase		Ordered liquid phase	
	Particle motion	Alignment	Linear instability	Basic object	Global pattern	Global order	GNF exponent
Polar/Vicsek	Polar	Ferro		Quantized traveling band	Smectic/irregular bands ^b	LRO	≥1.6
Active nematics ^c	Apolar	Nema	⊥	Macroscopic band ^d	Band chaos	QLRO	≥1.6

吸收态相变和向列有序耦合，密度涨落会是什么样的模式？

超均匀

巨涨落

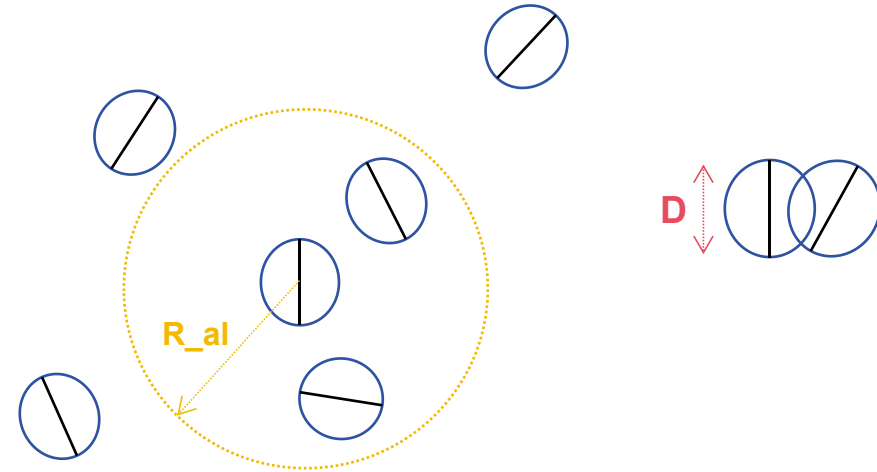
模型介绍--Nematic random organization model

- 运动规则

$$\theta_i^{t+1} = \frac{1}{2} \text{Arg} \left[\sum_{k \in \mathcal{V}_i} e^{i 2 \theta_k^t} \right] + \psi_i^t \pmod{\pi},$$

$$\mathbf{r}_i^{t+1} = \mathbf{r}_i^t \pm \delta_0 \mathbf{n}_i^t \quad (\text{overlapping})$$

$$\psi_i^t \in [-\sigma\pi/2, \sigma\pi/2] \quad \sigma \in [0, 1]$$



极限情况: $\sigma = 1$ 回到random organization model

$D \rightarrow \infty$ 回到nematic vicsek model

控制参量和序参量

$$\theta_i^{t+1} = \frac{1}{2} \operatorname{Arg} \left[\sum_{k \in \mathcal{V}_i} e^{i 2 \theta_k^t} \right] + \psi_i^t \pmod{\pi},$$

$$\mathbf{r}_i^{t+1} = \mathbf{r}_i^t \pm \delta_0 \mathbf{n}_i^t \quad (\text{overlapping})$$

- 控制参量 序参量
ROM: δ_0 $\phi = \frac{N\pi D^2}{4L^2}$ A
NVM: σ $\rho = \frac{NR_{\text{al}}^2}{L^2}$ $S = 2\sqrt{|\det Q|}$ $Q = N^{-1} \sum_i \mathbf{n}_i^t \mathbf{n}_i^t - \frac{1}{2} \mathbf{1}$
- 感兴趣的是活性有序和吸收态相变耦合后的密度涨落模式
选取参数 $\sigma = 0.1, \delta_0 = 0.3, R_{\text{al}} = 1, D = 0.46$
- 吸收态相变临界密度 $\phi_c \approx 0.30984$
 $\beta \approx 0.63$ β 符合CDP普适类

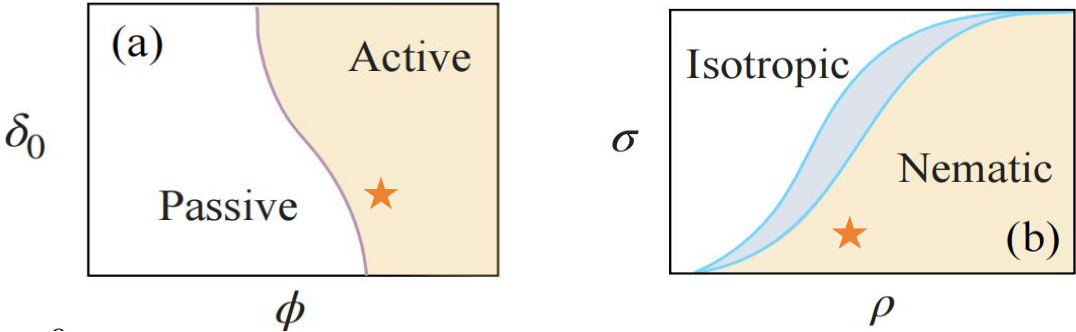


图1

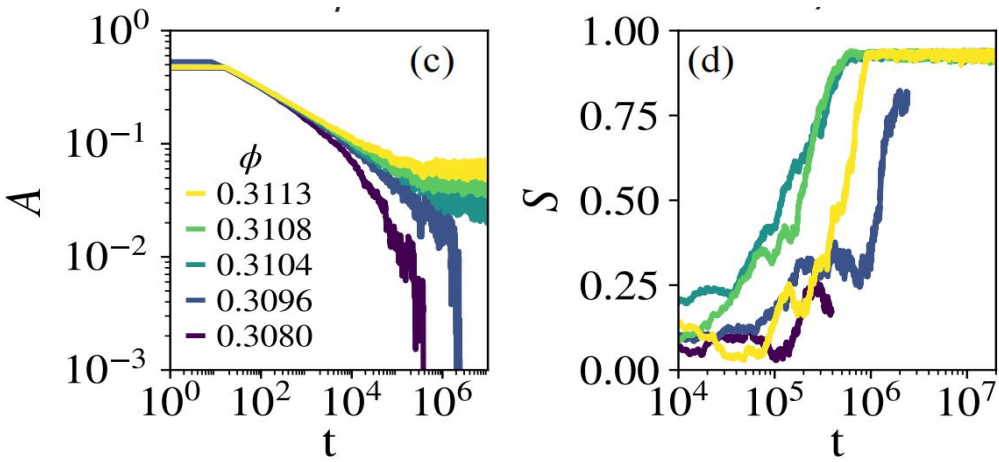


图2

CDP普适类	β	$\beta/\nu_{\perp}$	z	θ
	0.64(1)	0.78(2)	1.57(4)	0.42(1)

密度涨落模式

- **粒子数涨落** $\langle \Delta n^2 \rangle_\ell \sim \langle n \rangle_\ell^\alpha$

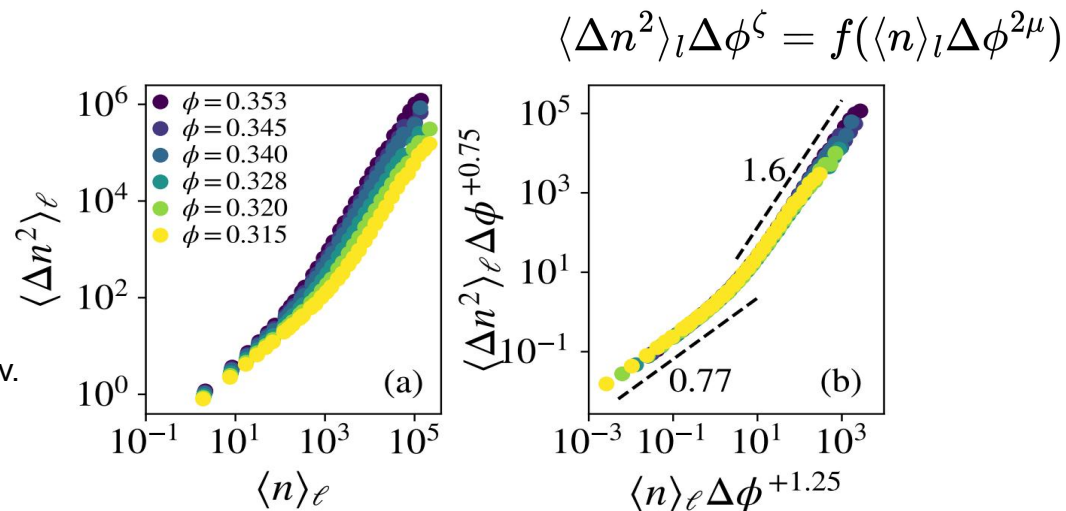
中间尺度 $\alpha = \alpha_{\text{hu}} \approx 0.77$ (2D ROM)

D. Hexner and D. Levine, Phys. Rev. Lett. **114**, 110602 (2015).

大尺度 $\alpha = \alpha_{\text{gnf}} \approx 1.6$ (2D nematic Vicsek model)

S. Ngo, A. Peshkov, I. S. Aranson, E. Bertin, F. Ginelli, and H. Chat'e, Phys. Rev. Lett. **113**, 038302 (2014).

crossover length $\xi \sim \Delta\phi^{-\mu}$ $\mu = 0.625$



- **结构因子** $S_{\text{iso}}(q)$

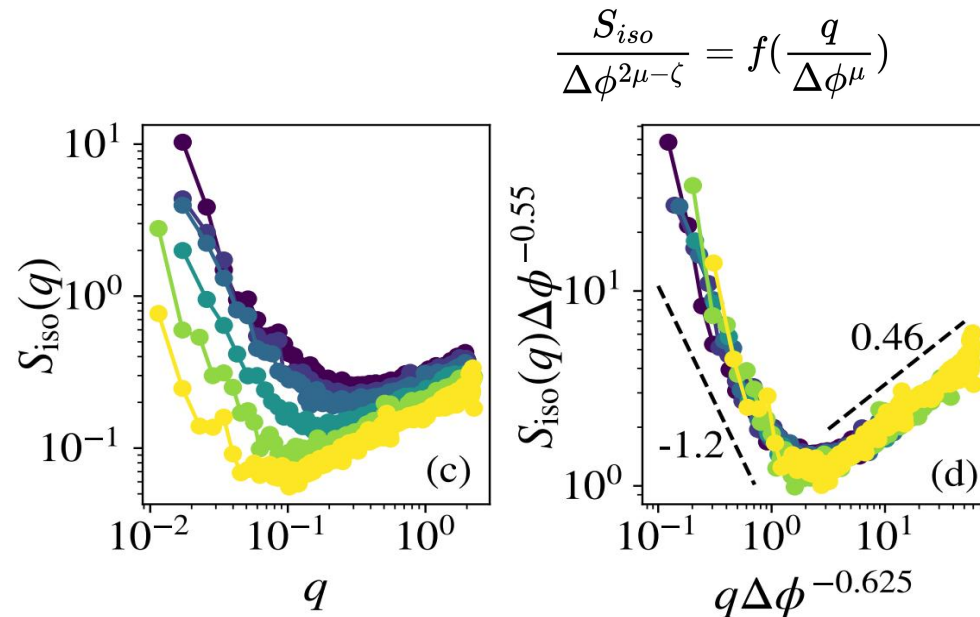
对 $S(\mathbf{q})$ 角度平均 $S(q) = N^{-1} \langle \sum_{i,j} e^{i\mathbf{q} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \rangle$

$S_{\text{iso}}(q) \sim q^\lambda$

$\lambda = (1 - \alpha)d$

$\lambda_{\text{hu}} \approx 0.46$ 超均匀

$\lambda_{\text{gnf}} \approx -1.2$ 巨涨落



反应项

$$\text{活性标量场} \quad \partial_t A = \nabla \cdot \mathbf{J} + \underbrace{(\kappa\rho - a)A - \lambda A^2}_{\text{反应项}} + \sigma_A \sqrt{A} \eta_A$$

$$\text{密度标量场} \quad \partial_t \rho = \underbrace{\nabla \cdot \mathbf{J} + \sigma_\rho \nabla \cdot (\sqrt{A} \eta_\rho)}_{\text{守恒场}}$$

$$\text{向列张量场} \quad \partial_t \mathbf{Q} = \underbrace{(\tilde{\mu}(\rho) - \gamma |\mathbf{Q}|^2) \mathbf{Q}}_{\text{对齐, Landau-like}} + D_Q \nabla^2 \mathbf{Q} + \chi_3 \widehat{\nabla \nabla} A + \sigma_Q \eta_Q$$

对齐, Landau-like

$$\mathbf{J} = D_\rho \nabla A + \chi_1 \mathbf{Q} \cdot \nabla A + \chi_2 A \nabla \cdot \mathbf{Q} \quad \eta \text{ 高斯噪声场}$$

- 无噪声, 吸收态相变临界密度 $\rho_0^c = a/\kappa$; 有噪声, 临界密度会低于 a/κ
- 如果 $\mu_0 = \tilde{\mu}(\rho_0) > 0$, 有序向列相

流体力学理论

- 假设有解

$$\begin{aligned}\rho &= \rho_0 \\ A_0 &= \frac{\kappa}{\lambda}(\rho_0 - \rho_c) \propto \Delta\phi \quad \text{with} \quad \rho_c = a/\kappa \\ S_0 &= \sqrt{\frac{2\mu_0}{\gamma}} > 0\end{aligned}$$

- 一阶微扰, 线性分析

$$\begin{aligned}\delta\rho &\equiv \rho(r, t) - \rho_0, \\ \delta A &\equiv A(r, t) - A_0, \\ \delta n_{\perp} &\equiv n_{\perp} \cdot (n(r, t) - n_0)\end{aligned}$$

- 进行时空傅里叶变换,

$$\tilde{\mathcal{F}}(\mathbf{q}, \omega) = \int dt \int d\mathbf{r} \mathcal{F}(\mathbf{r}, t) \exp[-i\mathbf{r} \cdot \mathbf{q} + i\omega t]$$

- 求解三元一次方程组,

$$(-i\omega \mathbf{1} - \mathbf{M}) \delta \mathbf{S} = \mathbf{b}$$

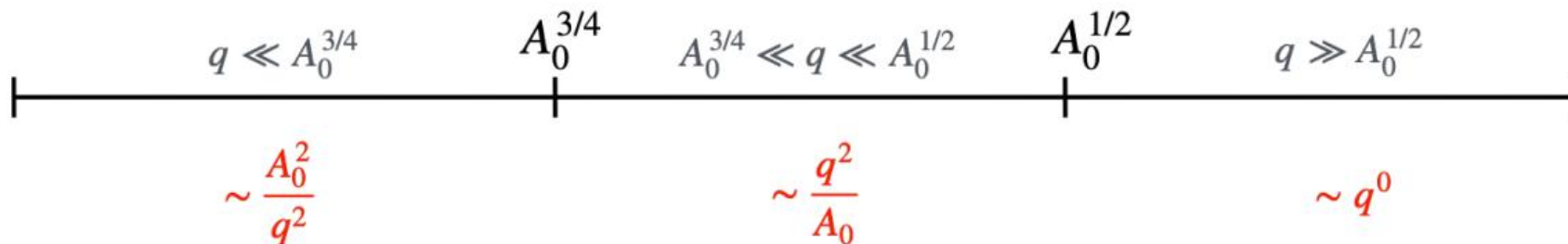
$$\delta \mathbf{S} = (\delta \tilde{A}, \delta \tilde{\rho}, \delta \tilde{n}_{\perp})^{\top}, \mathbf{b} = (\sigma_A \sqrt{A_0} \tilde{\eta}_A, iq\sigma_{\rho} \sqrt{A_0} \hat{\mathbf{q}} \cdot \tilde{\boldsymbol{\eta}}_{\rho}, \sigma_Q \tilde{\eta}_Q^{\perp}/S_0)^{\top}$$

$$\delta S_i = \frac{\det(-i\omega \mathbf{1} - \mathbf{M})|_{i \rightarrow b}}{\det(-i\omega \mathbf{1} - \mathbf{M})}$$

- 等时傅里叶变换

$$\begin{aligned}S(\mathbf{q}) &= \langle \delta\rho(\mathbf{q}, t) \delta\rho^*(\mathbf{q}', t) \rangle = \\ &= \frac{1}{(2\pi)^2} \int d\omega \int d\omega' \langle \delta\rho(\mathbf{q}, \omega) \delta\rho^*(\mathbf{q}', \omega') \rangle e^{-i(\omega + \omega')t}\end{aligned}$$

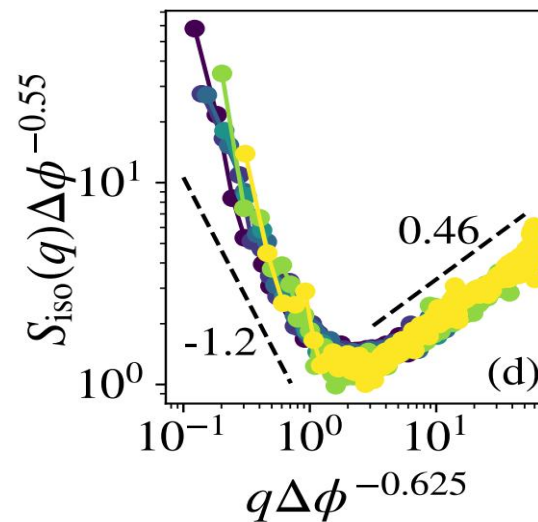
- 结果



$$\frac{S(q, \theta)}{A_0^{1/2}} \propto \frac{(1 - \cos(4\theta))\lambda^2\chi_2^2\sigma_Q^2}{D_Q D_\rho \kappa D_\rho + \lambda D_Q} \frac{1}{\tilde{q}^2} + \frac{2D_\rho\sigma_A^2}{\kappa\lambda} \tilde{q}^2 \quad \text{with} \quad \tilde{q} = qA_0^{-3/4} \quad \text{and} \quad A_0 \ll 1$$

$$q \ll A_0^{3/4}, \quad S(q) \sim q^2 \quad \text{巨涨落}$$

$$q \gg A_0^{3/4}, \quad S(q) \sim q^{-2} \quad \text{超均匀}$$



minimal theory

- 令 $\chi_1 = \chi_3 = 0, \gamma \gg 1, q \ll A_0^{1/2}$ 长时极限

$$\begin{aligned}
 -i\omega\delta\tilde{A}(\mathbf{q}, \omega) &= -q^2 D_\rho \delta\tilde{A} \left[-\lambda A_0 \delta\tilde{A} + \kappa A_0 \delta\tilde{\rho} \right] - \chi_2 A_0 S_0 q^2 \sin 2\theta \delta\tilde{\eta}_\perp + \sigma_A \sqrt{A_0} \tilde{\eta}_A \\
 -i\omega\delta\tilde{\rho}(\mathbf{q}, \omega) &= \left[-q^2 D_\rho \delta\tilde{A} \right] \left[-\chi_2 A_0 S_0 q^2 \sin 2\theta \delta\tilde{\eta}_\perp \right] + iq\sigma_\rho \sqrt{A_0} \hat{q} \cdot \tilde{\eta}_\rho \\
 -i\omega\delta\tilde{n}_\perp(\mathbf{q}, \omega) &= \left[-q^2 D_Q \delta\tilde{n}_\perp \right] + \frac{\sigma_Q}{S_0} \tilde{\eta}_Q^\perp
 \end{aligned}$$



$$\partial_t \delta\hat{\rho} = -Dq^2 \delta\hat{\rho} + \underbrace{\frac{\tilde{\sigma}_A q^2}{\sqrt{A_0}} \hat{\eta}_A}_{\text{superconservative noise}} + \underbrace{\tilde{\sigma}_Q \sin 2\theta A_0 \hat{\eta}_Q^\perp}_{\text{nonconserved noise}}$$

superconservative noise

nonconserved noise

$$\propto \frac{q^4}{A_0}$$

$$\propto A_0^2$$

- 来自活性场的守恒噪声和来自向列横场的非守恒噪声的竞争

- 两噪声可比时 $q \sim A_0^{\frac{3}{4}}$

$$S(q) \sim \frac{q^2}{A_0} \quad q \gg A_0^{\frac{3}{4}}$$

守恒噪声，超均匀

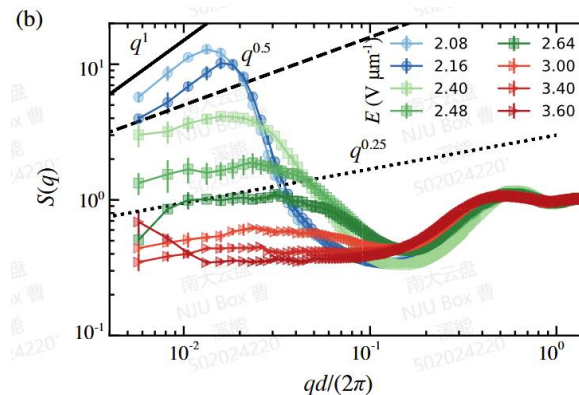
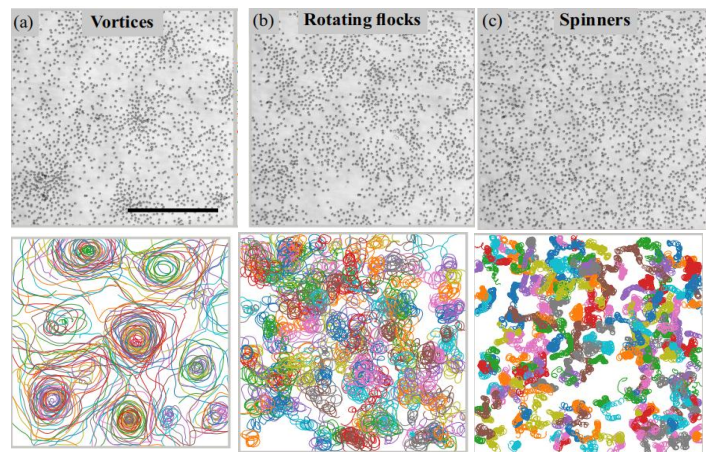
$$S(q) \sim (1 - \cos 4\theta) \frac{A_0^2}{q^2} \quad q \ll A_0^{\frac{3}{4}}$$

非守恒噪声，巨涨落

另一个现象

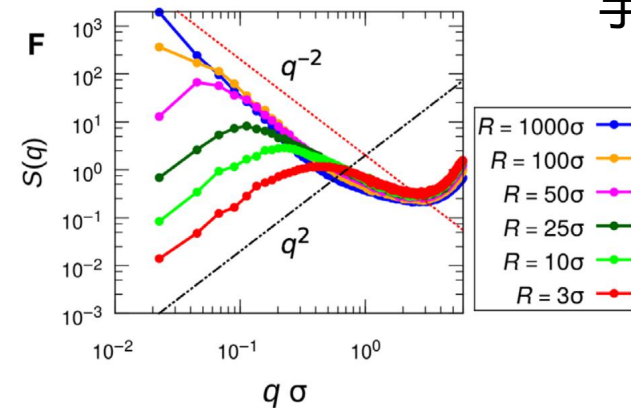
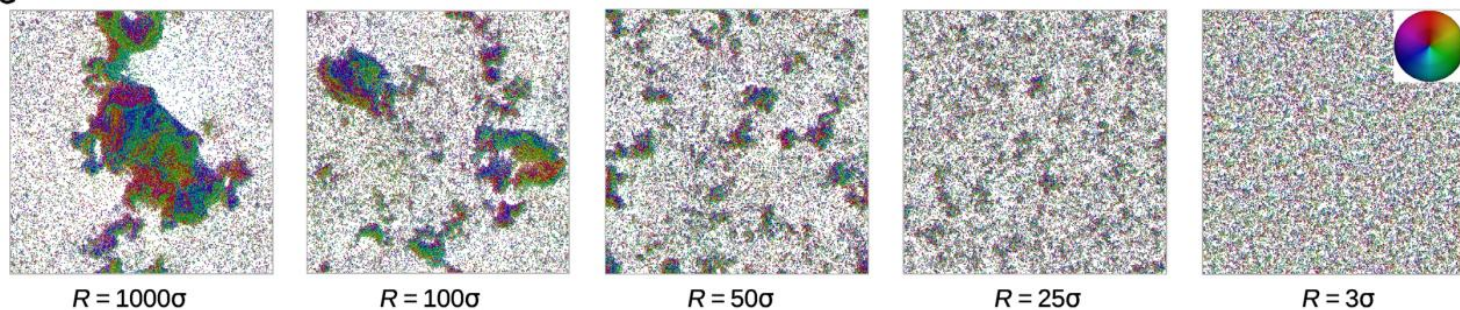
- 局部巨涨落，全局超均匀

Zhang B, Snezhko A. Hyperuniform active chiral fluids with tunable internal structure[J]. Physical Review Letters, 2022, 128(21): 218002.



梨形quincke roller

Lei Q L, Ciamarra M P, Ni R. Nonequilibrium strongly hyperuniform fluids of circle active particles with large local density fluctuations[J]. Science advances, 2019, 5(1): eaau7423.



手性活性粒子

总结

- 耦合活性，向列有序，吸收态相变的最小模型
- 不同异常密度涨落在不同尺度下共存。局部超均匀，全局巨涨落
- 来源于守恒噪声和非守恒噪声的竞争
- 另一种现象：局部巨涨落，全局超均匀